

OpenClaw Scheduled Tasks Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the "automatically execute scripts at fixed time" part.

1. Tutorial Objectives

This article demonstrates how OpenClaw can further upgrade from instant dialogue to automated peripheral applications through "AI-generated scripts + Linux timed execution". After completing this tutorial, readers will be able to:

- Understand the basic implementation approach for OpenClaw scheduled tasks
- Distinguish between scenarios suitable for `HEARTBEAT` and `cron` respectively
- Describe automation requirements via WhatsApp, TUI, and voice commands

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display screen
- RGB light strip / RGB LED module
- `[Optional] CSI camera`
- `[Optional] AI voice module + speaker`

Note:

- The core of this tutorial is "timed execution", not requiring all peripherals to be connected at the same time
- If you want to do which type of automation task, just connect the corresponding hardware first

3. Testing Scheduled Task Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, proceed with the scheduled task-specific test.

It is recommended to test in the following order:

1. First enter OpenClaw TUI
2. Have OpenClaw generate a simple script
3. Manually run the script in the terminal to confirm normal function
4. Then write the script to `crontab`

Input in terminal:

```
openclaw tui
```

It is recommended to first try these types of requirements:

- Help me write a script that displays current temperature and humidity on OLED and saves to desktop
- Generate a script that reads temperature and humidity and prints results each time it runs
- Write a script that makes RGB light up green for 3 seconds then turns off

After script generation, run it manually first, for example:

```
python3 /home/pi/Desktop/your_script.py
```

After confirming the script itself is usable, then execute:

```
crontab -e
crontab -l
```

4. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote automation requirement description	Suitable for first clarifying "what you want to schedule"	Used to generate scripts and confirm requirements
TUI	Local creation and debugging of scheduled tasks	Easiest to coordinate scripts and <code>cron</code>	First choice for implementation solution
Voice Code	Voice description of automation scenarios	More like desktop assistant experience	Suitable for demonstrating "generate automation from one sentence"

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

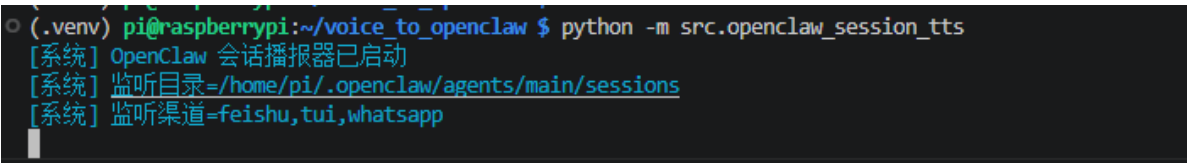
- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



4.1 WhatsApp Solution

For WhatsApp integration steps, please refer directly to [05-Three Dialogue Configuration Tests]. This section is more suitable for describing requirements, for example:

- Help me create a scheduled task: make RGB light up red every minute as a reminder, then turn off the light

20:42



< 2



+86 153 3885 7526 (你)

给自己发消息

Help me create a schedule task: make RGB lights up red every minute as a reminder, the turn off the light.

20:41 ✓✓

Schedule task created successfully!

Task Details:

- **Name:** RGB Red Minute Reminder
- **Schedule:** Every 60 seconds (1 minute)
- **Job**
ID: e13c213c-223e-4408-92f5-ef708d8d41ce
- **Status:** Enabled ✓
- **Next Run:** In ~60 seconds

What happens each minute:

1. 💡 RGB lights turn **RED** at 80% brightness
2. ⌚ Stay lit for 2 seconds as a reminder
3. 🚫 Lights turn off automatically

You'll receive a WhatsApp notification when the task runs. The red minute reminder is now active! 🕒🔴

20:42 ✓✓

RGB minute reminder executed successfully:

- Turned on all RGB LEDs (blue color 80% brightness)

Waited 2 seconds



```
openclaw tui
```

- Help me create a scheduled task: read temperature and humidity every minute and display on OLED

```
File Edit Tabs Help
Help me create a scheduled task: read temperature and humidity every minute
and display on OLED

Scheduled task created successfully! 📅

Task Details:
- Name: Temperature Humidity Dashboard
- Schedule: Every 1 minute (60,000ms)
- Job ID: a43df4b4-6c47-464e-8cce-95ed29bc8a36
- Status: Enabled ✓
- Next Run: In ~60 seconds

What it does:
- Reads temperature and humidity from the DHT11 sensor
- Displays the values on the OLED screen
- Runs automatically every minute in the main session

To manage this task:
- View all jobs: Use cron list command
- Remove job: Use cron remove with the job ID above
- Pause job: Use cron update to set enabled=false

The temperature dashboard will now update automatically every minute! 📅
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
82k/1.0m (8%)
```

Requires optional AI voice interaction module + speaker

- Help me create a scheduled task: read temperature and humidity every minute and display on OLED

- Help me create a scheduled task: read temperature and humidity every minute and display on OLED

```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Help me create a scheduled task, read temperature and humidity every minute
and display on aled.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## ✔ Scheduled Task Created!

**Task Details:**
- **Name:**
    Temperature Humidity Monitor
- **Job ID:**
    `1a6954fc-f6e9-4aed-8e2f-a5737df6072d`
- **Schedule:** Every 60
    seconds (1 minute)
- **Status:** Enabled ✔
- **Next Run:** In ~60
    seconds

**What It Does:**
1. Reads temperature and humidity from
    the DHT11 sensor
2. Displays the values on the OLED screen
3.
    Repeats automatically every minute

**To Manage This Task:**
-
```

5. Project Implementation

Note: All demonstrations below use the `openclaw_tui` method for conversation. You can also choose WhatsApp or voice to complete.

5.2 Scenario 1: Hourly Temperature and Humidity Sampling

Experiment Objective

Have the device automatically read temperature and humidity once every hour and record results or sync to the screen.

Effect Description

This is the most basic and common scheduled task scenario, which can continue to be extended into logs, dashboards, environment alerts, or plant care projects.

Example Commands

- Help me create a scheduled task: read temperature and humidity every minute and display on OLED

5.3 Scenario 2: Timed Status Reminder

Experiment Objective

Use RGB or OLED to make a clear reminder at a fixed time.

Effect Description

This type of task is suitable for water drinking reminders, class reminders, inspection reminders, and status switch reminders. It does not emphasize continuous operation but emphasizes triggering a clear action at a certain time point.

Example Commands

- Help me create a scheduled task: make RGB light up red every minute as a reminder, then turn off the light

5.4 Scenario 3: Timed Inspection Script

Experiment Objective

Chain multiple peripheral actions together to form a small automated task that can be executed periodically.

Effect Description

For example, read temperature and humidity once every hour. If the status is abnormal, light up to give a reminder, and display the results on OLED. This type of task is already close to a complete small project, and is also where the combination of "AI-generated scripts" and "timed execution" is most evident.

Example Commands

- Help me create a scheduled task: read temperature and humidity once every minute, light up red if abnormal, and prompt on OLED